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ATOMIZER

Abstract

An atomizer is provided. The atomizer includes: a mouthpiece, a first end of which being configured to be placed at a user's mouth; a delivery tube holder located at a second end of the mouthpiece opposite to the first end; a guide mechanism for guiding the delivery tube holder to move between a first position and a third position in an axial direction of the mouthpiece, the delivery tube holder is closest to the mouthpiece in the axial direction when it is in the third position, and the delivery tube holder is farthest from the mouthpiece in the axial direction when it is in the first position; the atomizer performs spraying during the movement of the delivery tube holder from the first position to the third position, and the atomizer performs liquid metering during the movement of the delivery tube holder from the third position to the first position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation of the international application PCT/CN2023/132423, filed on Nov. 17, 2023 and entitled "ATOMIZER", and the international application claims the priority to Chinese Patent Application No. 202211485092.8 filed on Nov. 24, 2022, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD
[0002] The disclosure relates to the technical field of medical instruments. Specifically, the disclosure relates to an atomizer which may be used, for example, for atomizing and/or spraying a medicinal liquid.
BACKGROUND
[0003] In the related art, a container in an atomizer (or referred to as a sprayer) contains a liquid to be atomized or sprayed, and during a movement stroke of the container relative to a spraying assembly (such as a pump), the liquid in the container is atomized, and the atomized liquid is sprayed from a nozzle of the spraying assembly.
SUMMARY OF THE INVENTION
[0004] The disclosure provides an atomizer.
[0005] According to one aspect of the disclosure, an atomizer is provided. The atomizer includes: a mouthpiece, a first end of the mouthpiece being configured to be placed at a mouth of a user; a delivery tube holder located at a second end of the mouthpiece opposite to the first end; and a guide mechanism configured to guide the delivery tube holder to move between a first position and a third position in an axial direction of the mouthpiece, where the delivery tube holder is closest to the mouthpiece in the axial direction when the delivery tube holder is in the third position, and the delivery tube holder is farthest from the mouthpiece in the axial direction when the delivery tube holder is in the first position; and where the atomizer performs spraying during the movement of the delivery tube holder from the first position to the third position, and the atomizer performs liquid metering during the movement of the delivery tube holder from the third position to the first position.
[0006] According to some embodiments, there is a second position between the first position and the third position, the guide mechanism is configured to guide the delivery tube holder to move between the first position, the second position and the third position in the axial direction of the mouthpiece, the atomizer performs a first spraying operation during the movement of the delivery tube holder from the first position to the second position, and the atomizer performs a second spraying operation during the movement of the delivery tube holder from the second position to the third position.
[0007] According to some embodiments, the guide mechanism includes: a first guide structure arranged on an inner wall of the delivery tube holder, with a surface of the first guide structure facing the mouthpiece forming a first guide surface; and a second guide structure arranged on an inner wall of the mouthpiece, with a surface of the second guide structure facing the delivery tube holder forming a second guide surface.
[0008] According to some embodiments, when a first portion of the first guide surface abuts on a

second portion of the second guide surface, the delivery tube holder is in the first position; when a third portion of the first guide surface abuts on a fourth portion of the second guide surface, the delivery tube holder is in the second position; and when the first guide surface is separated from the second guide surface, the delivery tube holder is in the third position.

[0009] According to some embodiments, the first guide surface includes: a guide portion configured to guide the delivery tube holder to move from the third position to the first position; and a stop portion configured to guide the delivery tube holder to move from the first position to the second position.

[0010] According to some embodiments, the guide portion is an inclined surface, and the stop portion is a step surface and includes an axial flat portion parallel to the axial direction and a radial flat portion parallel to a radial direction of the delivery tube holder.

[0011] According to some embodiments, the first portion is the highest point of the guide portion in the axial direction, and the second portion is the highest point of the second guide surface in the axial direction; and the third portion is the radial flat portion of the stop portion, and the fourth portion is the same portion as the second portion.

[0012] According to some embodiments, the first guide surface further includes a transition portion located between the guide portion and the stop portion and configured as a cambered surface.

[0013] According to some embodiments, the width of the first guide surface in a radial direction of the delivery tube holder is different from the width of the second guide surface in the radial direction.

[0014] According to some embodiments, the first guide structure is integrally formed with the delivery tube holder, and the second guide structure is integrally formed with the mouthpiece.

[0015] According to some embodiments, the atomizer further includes an actuating member, where the actuating member is capable of being actuated to drive the delivery tube holder to pivot relative to the mouthpiece, and the pivoted delivery tube holder is capable of being guided by the guide mechanism to move in the axial direction.

[0016] According to some embodiments, the atomizer further includes a locking mechanism, where the locking mechanism switches the actuating member between a locked position, in which the mouthpiece stops the actuating member from pivoting, and an unlocked position, in which the actuating member is capable of pivoting relative to the mouthpiece.

[0017] According to some embodiments, when the delivery tube holder is in the first position and the third position, the actuating member is in the unlocked position; and when the delivery tube holder is in the second position, the actuating member is in the locked position.

[0018] According to some embodiments, the atomizer further includes a switch button, where in the locked position, the switch bottom is capable of being pressed to drive the delivery tube holder to pivot to separate the first guide surface from the second guide surface.

[0019] According to some embodiments, the atomizer further includes a bottom spring arranged at the bottom of the delivery tube holder, where the bottom spring is configured to apply a thrust to the delivery tube holder to drive the delivery tube holder to move toward the mouthpiece, and when the first guide surface is separated from the second guide surface, the delivery tube holder is capable of moving from the second position to the third position under the action of the thrust of the bottom spring.

[0020] According to some embodiments, the locking mechanism includes: a locking slider provided on the mouthpiece; a slot provided in a circumferential wall of the actuating member; and a protrusion provided on a circumferential surface of the delivery tube holder and extending into the slot, where when the delivery tube holder is in the first position, the locking slider abuts against an end surface of the actuating member facing the mouthpiece, such that the actuating member is in the unlocked position; where when the delivery tube holder is in the second position, a portion of the locking slider is inserted into the slot and abuts on the protrusion, such that the actuating member is in the locked position; and where when the delivery tube holder is in the third position,

the locking slider is pushed by the protrusion to return to the mouthpiece, such that the actuating member is in the unlocked position.
[0021] According to some embodiments, when the delivery tube holder is in the first position, the atomizer is in a pre-cleaning state; when the delivery tube holder is in the second position, the atomizer is in a cleaned state and a pre-triggering state; and when the delivery tube holder is in the third position, the atomizer is in a triggered state or an initial state.
[0022] According to some embodiments, when the delivery tube holder is in the first position, the actuating member is in the locked position; and when the delivery tube holder is in the third position, the actuating member is in the unlocked position.
[0023] According to some embodiments, the locking mechanism includes: a locking slider provided on the mouthpiece; a slot provided in a circumferential wall of the actuating member; and a protrusion provided on a circumferential surface of the delivery tube holder and extending into the slot, where when the delivery tube holder is in the first position, a portion of the locking slider is inserted into the slot and abuts on the protrusion, such that the actuating member is in the locked position; and where when the delivery tube holder is in the third position, the locking slider is pushed by the protrusion to return to the mouthpiece, such that the actuating member is in the unlocked position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0024] To describe technical solutions in embodiments of the disclosure or in the prior art more clearly, accompanying drawings required for description of the embodiments or the prior art will be briefly introduced below. Apparently, the accompanying drawings in the following description show some embodiments of the disclosure merely, and those of ordinary skill in the art can derive other drawings from structures shown in these accompanying drawings without any creative effort.
[0025] In the figures:
[0026] FIG. **1** shows a front view of an atomizer according to some embodiments of the disclosure;
[0027] FIG. **2** shows a cross-sectional view of the atomizer in FIG. **1** taken along R-R;
[0028] FIG. **3** shows an exploded view of the atomizer in FIG. **1**;
[0029] FIGS. **4**A to **4**D show schematic diagrams of the atomizer in FIG. **1** in a pre-cleaning state from different perspectives, FIG. **4**A showing a cross-sectional view of the atomizer in FIG. **1** in the pre-cleaning state taken along N-N, FIG. **4**B showing a schematic diagram of some components of the atomizer in the pre-cleaning state, FIG. **4**C showing a cross-sectional view of some components in FIG. **4**B taken along K-K, and FIG. **4**D showing an enlarged view of an area H of the atomizer in FIG. **4**A;
[0030] FIGS. **5**A to **5**D show schematic diagrams of the atomizer in FIG. **1** in a cleaned state from different perspectives, FIG. **5**A showing a cross-sectional view of the atomizer in FIG. **1** in the cleaned state taken along N-N, FIG. **5**B showing a schematic diagram of some components of the atomizer in the cleaned state, FIG. **5**C showing a cross-sectional view of some components in FIG. **5**B taken along D-D, and FIG. **5**D showing an enlarged view of an area P of the atomizer in FIG. **5**A;
[0031] FIGS. **6**A to **6**D show schematic diagrams of the atomizer in FIG. **1** in a triggered state from different perspectives, FIG. **6**A showing a cross-sectional view of the atomizer in FIG. **1** in the triggered state taken along N-N, FIG. **6**B showing a schematic diagram of some components of the atomizer in the triggered state, FIG. **6**C showing a cross-sectional view of some components in FIG. **6**B taken along E-E, and FIG. **6**D showing an enlarged view of an area Q of the atomizer in FIG. **6**A;
[0032] FIG. **7**A shows a schematic diagram of a delivery tube holder of the atomizer in FIG. **1**;

[0033] FIG. **7**B shows a top view of the delivery tube holder in FIG. **7**A;
[0034] FIG. **8**A shows a schematic diagram of a mouthpiece of the atomizer in FIG. **1**;
[0035] FIG. **8**B shows a top view of the mouthpiece in FIG. **8**A;
[0036] FIG. **9** shows a schematic diagram of a locking slider of the atomizer in FIG. **1**;
[0037] FIG. **10** shows a schematic diagram of an actuating member of the atomizer in FIG. **1**;
[0038] FIG. **11** shows a schematic diagram of a switch button of the atomizer in FIG. **1**; and
[0039] FIGS. **12**A to **12**C show schematic diagrams of an atomizer according to some other embodiments of the disclosure in a pre-triggering state from different perspectives, FIG. **12**A showing a cross-sectional view of the atomizer in the pre-triggering state taken along a vertical direction, FIG. **12**B showing a schematic diagram of some components of the atomizer in the pre-triggering state, and FIG. **12**C showing a cross-sectional view of some components in FIG. **12**B taken along D′-D′; and
[0040] FIGS. **13**A to **13**C show schematic diagrams of the atomizer in FIG. **12**A in a triggered state from different perspectives, FIG. **13**A showing a cross-sectional view of the atomizer in the triggered state taken along a vertical direction, FIG. **13**B showing a schematic diagram of some components of the atomizer in the triggered state, and FIG. **13**C showing a cross-sectional view of some components in FIG. **13**B taken along E′-E′.
[0041] In the figures, the same reference signs refer to the same or similar features.
[0042] The implementation of the objective, functional characteristics, and advantages of the disclosure will be further described with reference to the accompanying drawings in combination with embodiments.

DETAILED DESCRIPTION

[0043] In the scope of the disclosure, an “atomizer” (also referred to as a “sprayer”) refers to a device for atomizing a liquid. Generally, the atomizer is configured to atomize a fluid (e.g., a medicinal liquid or the like) and spray the atomized fluid to some parts of a user (such as a patient) to be treated. Since the medicinal liquid is loaded in the atomizer, the stability of the atomizer is particularly important.
[0044] In the related art, the atomizer is usually susceptible to contaminants in the environment, residual medicinal liquid from the previous spraying, and a medicinal liquid volatilized into a spraying assembly before formal spraying, resulting in unstable drug delivery dose from the atomizer and thus affecting the efficacy of the medicinal liquid.
[0045] In the disclosure, a delivery tube holder is guided by a guide mechanism to move between a first position, a second position and a third position, so as to perform two spraying operations, that is, a first spraying operation (pre-spraying) and a second spraying operation (formal drug-delivery spraying) within one stroke cycle. The pre-spraying can prevent the residual medicinal liquid in the spraying assembly from mixing in the formal drug-delivery spraying and thus stabilize the dose of the medicinal liquid, and can also remove contaminants at the nozzle of a nozzle assembly to prevent the contaminants from being inhaled with the drug and thus achieve a cleaning effect. This allows for safer, more stable, more reliable and more effective drug delivery.
[0046] In the scope of the disclosure, the “locked position” of the actuating member refers to a position in which the actuating member is locked, that is, the atomizer cannot be triggered without the action of other components. In other words, when the actuating member is in the “locked position”, the atomizer can perform an atomization or spraying operation only manually by means of other components.
[0047] In the scope of the disclosure, the “unlocked position” of the actuating member refers to a position in which the actuating member is not locked, that is, a position in which the atomizer can be operated by means of the actuating member. In other words, when the actuating member is in the “unlocked position”, a user can rotate the actuating member to drive the delivery tube holder to perform corresponding rotation and axial movement at the same time.
[0048] In the scope of the disclosure, the “initial state” of the atomizer refers to a state in which no

operation is performed on the atomizer, the "pre-cleaning state" refers to a critical state of the atomizer before pre-spraying, the "cleaned state" refers to a state in which the atomizer has completed the pre-spraying, that is, a pre-spraying operation has just been completed and no other operations are performed, the "pre-triggering state" refers to a critical state of the atomizer before formal drug-delivery spraying, and the "triggered state" refers to a state in which the atomizer has completed the formal drug-delivery spraying, that is the spraying has just been completed and no other operations are performed. In some examples, the "cleaned state" and the "pre-triggering state" are the same state, specifically, the components in the atomizer are in the same positions and states.

[0049] An atomizer according to some specific embodiments of the disclosure will be described in detail below with reference to the embodiments shown in FIGS. **1** to **13**.

[0050] As shown in FIGS. **1**, **2** and **3**, an atomizer **1000** may include a mouthpiece **400**, a delivery tube holder **300** and a guide mechanism **440**. The guide mechanism **440** is configured to guide the delivery tube holder **300** to move between a first position (as shown in FIG. **4**A) and a third position (as shown in FIG. **6**A) in an axial direction L of the mouthpiece **400**. The delivery tube holder **300** is closest to the mouthpiece **400** in the axial direction L when the delivery tube holder is in the third position, and the delivery tube holder **300** is farthest from the mouthpiece **400** in the axial direction L when the delivery tube holder is in the first position. The atomizer **1000** performs spraying during the movement of the delivery tube holder **300** from the first position to the third position, and the atomizer **1000** performs liquid metering during the movement of the delivery tube holder **300** from the third position to the first position. Liquid metering herein refers to a liquid being drawn into a metering chamber of a spraying assembly **800**, such that the liquid (i.e., a medicinal liquid) is ready before the spraying.

[0051] In some embodiments, as shown in FIGS. **12**A to **13**C, the guide mechanism **440′** may be configured to guide the delivery tube holder **300** to move between the first position (as shown in FIG. **12**A) and the third position (as shown in FIG. **13**A) in the axial direction L of the mouthpiece **400**, so as to achieve spraying of the atomizer. That is, as shown in FIG. **12**A, when the delivery tube holder **300** is in the first position, the delivery tube holder **300** is in the lowest position in the axial direction L. As shown in FIG. **13**A, when the delivery tube holder **300** is in the third position, the delivery tube holder **300** is in the highest position in the axial direction L, that is, the delivery tube holder moves a distance upward from the first position in the axial direction. Thus, during the movement stroke of the delivery tube holder **300** from the first position to the third position, the liquid in a container **700** is atomized and spayed from a nozzle of the spraying assembly **800**, such that a spraying operation (i.e., the formal drug-delivery spraying) is performed. During the movement from the third position to the first position, the liquid in the container **700** is dawn into the metering chamber of the spraying assembly **800** and can thus be reused.

[0052] In some embodiments, as shown in FIGS. **4**A to **6**D, there is a second position between the first position and the third position. In this case, the guide mechanism **440** may be configured to guide the delivery tube holder **300** to move between the first position (as shown in FIG. **4**A), the second position (as shown in FIG. **5**A) and the third position (as shown in FIG. **6**A) in the axial direction L of the mouthpiece **400**. The atomizer **1000** performs liquid metering during the movement of the delivery tube holder **300** from the third position to the first position, the atomizer **1000** performs a first spraying operation during the movement of the delivery tube holder **300** from the first position to the second position, and the atomizer **1000** performs a second spraying operation during the movement of the delivery tube holder **300** from the second position to the third position, so as to complete one dosing cycle.

[0053] That is, as shown in FIG. **4**A, when the delivery tube holder **300** is in the first position, the delivery tube holder **300** is in the lowest position in the axial direction L. As shown in FIG. **5**A, when the delivery tube holder **300** is in the second position, the delivery tube holder **300** is in an intermediate position between the highest position and the lowest position in the axial direction L, that is, the delivery tube holder moves a distance upward from the first position in the axial direction. As shown in FIG. **6**A, when the delivery tube holder **300** is in the third position, the delivery tube holder **300** is in the highest position in the axial direction L, that is, the delivery tube holder moves a distance upward from the second position in the axial direction. Thus, during a movement stroke of the delivery tube holder **300** from the first position to the second position, the liquid in a container **700** connected to the delivery tube holder **300** is atomized and spayed from a nozzle of the spraying assembly **800**, such that the first spraying operation (i.e., the pre-spraying) is performed. During a movement stroke of the delivery tube holder **300** from the second position to the third position, the liquid in the container **700** is atomized and spayed from the nozzle of the spraying assembly **800**, such that the second spraying operation (i.e., the formal drug-delivery spraying) is performed. During the movement from the third position to the first position, the liquid in the container **700** is dawn into the metering chamber of the spraying assembly **800** and can thus be reused. The atomizer **1000** is guided by the guide mechanism **440** to move between the first position, the second position and the third position, so as to perform two spraying operations, that is, the pre-spraying and the formal spraying, within one stroke cycle. The pre-spraying before the formal spraying can prevent the residual medicinal liquid in the spraying assembly **800** from mixing in the formal drug-delivery spraying and thus stabilize the dose of the medicinal liquid, and can also remove contaminants at the nozzle of a nozzle assembly to prevent the contaminants from being inhaled with the drug and thus achieve a cleaning effect. This allows for safer, more stable, more reliable and more effective drug delivery.

[0054] An atomizer **1000** according to some specific embodiments of the disclosure in which pre-spraying is performed will be described in detail below with reference to the embodiments shown in FIGS. **1** to **11**.

[0055] In some embodiments, when the delivery tube holder **300** is in the first position, the atomizer **1000** is in the pre-cleaning state, as shown in FIG. **4**A. When the delivery tube holder **300** is in the second position, the atomizer **1000** is in the cleaned state and the pre-triggering state, as shown in FIG. **5**A. When the delivery tube holder **300** is in the third position, the atomizer **1000** is in the triggered state or the initial state, as shown in FIG. **6**A. Thus, the atomizer **1000** is guided by the guide mechanism **440** to perform two spraying operations, that is, the pre-spraying and the formal drug-delivery spraying, within one stroke cycle.

[0056] In some embodiments, as shown in FIGS. **4**D, **5**D and **6**D, the guide mechanism **440** may include: a first guide structure **330** arranged on an inner wall of the delivery tube holder **300**, with a surface of the first guide structure **330** facing the mouthpiece **400** forming a first guide surface **331**; and a second guide structure **430** arranged on an inner wall of the mouthpiece **400** (specifically, a mouthpiece body **410**), with a surface of the second guide structure **430** facing the delivery tube holder **300** forming a second guide surface **431**. Thus, during the pivoting of the delivery tube holder **300**, the delivery tube holder **300** can move in the axial direction under the interaction of the first guide surface **331** and the second guide surface **431** that are in contact with each other, that is, can perform a composite motion of rotation and axial translation, so as to change the position of the delivery tube holder **300** in the axial direction L.

[0057] In some embodiments, during the pivoting of the delivery tube holder **300**, the delivery tube holder **300** may move downward while rotating under the action of the first guide surface **331** and the second guide surface **431**, and during the upward movement of the delivery tube holder **300**, the delivery tube holder **300** may perform a simple upward translation without any rotation. The disclosure is not limited thereto.

[0058] In some embodiments, as shown in FIGS. **4**D, **5**D and **6**D, when a first portion of the first guide surface **331** abuts on a second portion of the second guide surface **431**, the delivery tube holder **300** is in the first position; when a third portion of the first guide surface **331** abuts on a fourth portion of the second guide surface **431**, the delivery tube holder **300** is in the second position; and when the first guide surface **331** is separated from the second guide surface **431**, the delivery tube holder **300** is in the third position. That is, the delivery tube holder **300** may be guided by the first guide surface **331** and the second guide surface **431** to be in the first position, is stopped by the first guide surface **331** and the second guide surface **431** to be in the second position, and then, after the first guide surface **331** is separated from the second guide surface **431** (i.e., having no guiding effect), enters the third position under the action of, for example, an upward thrust of a bottom spring **600** of the delivery tube holder **300**. Thus, the delivery tube holder **300** can move from the first position to the second position and then to the third position in the axial direction L by means of the guide mechanism **440**. Alternatively, it may also be configured that when a fifth portion of the first guide surface **331** abuts on a sixth portion of the second guide surface **431**, the delivery tube holder **300** is in the third position. The disclosure is not limited thereto.

[0059] In some embodiments, as shown in FIGS. **4**D, **5**D, **6**D, **7**A and **7**B, the first guide surface **331** of the first guide structure **330** on the delivery tube holder **300** may include: a guide portion **333** configured to guide the delivery tube holder **300** to move from the third position to the first position; and a stop portion **332** configured to guide the delivery tube holder **300** to move from the first position to the second position and stop the delivery tube holder **300** from moving from the second position to the first position. Thus, the guide mechanism **440** can guide the delivery tube holder **300** to move from top to bottom in the axial direction L, and stop the delivery tube holder **300** from moving from bottom to top in the axial direction L, such that the pre-spraying is performed before the formal spraying.

[0060] In some embodiments, as shown in FIGS. **7**A and **7**B, the guide portion **333** is an inclined surface, the stop portion **332** is a step surface. The stop portion **332** may include an axial flat portion **332-2** parallel to the axial direction L and a radial flat portion **332-1** parallel to a radial direction J of the delivery tube holder **300**. The inclined surface may be a spiral surface, for example, a spiral surface that spirally extends upward from an inner bottom wall of the delivery tube holder **300** around a central axis of the delivery tube holder **300**. Alternatively, the guide portion **333** may also be an inclined surface or a curved surface that extends in any direction and has a height gradually increasing from the inner bottom wall of the delivery tube holder **300**. The disclosure is not limited to the example described above. Alternatively, the step surface may also include a flat portion or a curved portion extending in a direction at an angle relative to the axial direction L, and a flat portion or a curved portion extending in a direction at an angle relative to the radial direction J of the delivery tube holder **300**. The disclosure is not limited to the example described above. Thus, the guide mechanism **440** can guide the delivery tube holder **300** to move from top to bottom in the axial direction L, and stop the delivery tube holder **300** from moving from bottom to top in the axial direction L, such that the pre-spraying is performed before the formal drug-delivery spraying.

[0061] In some embodiments, as shown in FIGS. **8**A and **8**B, the second guide surface **431** of the second guide structure **430** on the mouthpiece **400** may include an inclined portion **432** and a flat portion **433**. The inclined portion **432** may be a spiral surface, for example, a spiral surface extending upward from an inner bottom wall of the mouthpiece **400** around a central axis of the mouthpiece **400**. Alternatively, the guide portion **333** may also be an inclined surface or a curved surface that extends in any direction and has a height gradually increasing from the inner bottom wall of the mouthpiece **400**. The disclosure is not limited to the example described above. The flat portion may be in the highest position of the second guide surface **431** in the axial direction L.

[0062] In some other embodiments, the second guide surface **431** of the second guide structure **430** on the mouthpiece **400** may include a guide portion configured to guide the delivery tube holder **300** to move from the third position to the first position and a stop portion configured to guide the delivery tube holder **300** to move from the first position to the second position. Correspondingly, the first guide surface **331** of the first guide structure **330** on the delivery tube holder **300** may only include an inclined portion and a flat portion.

[0063] In some embodiments, as shown in FIGS. **4**D and **5**D, the first portion is the highest point of the guide portion **333** of the first guide surface **331** in the axial direction L, and the second portion is the highest point of the second guide surface **431** in the axial direction L. The third portion is the radial flat portion **332-1** of the stop portion **332**, and the fourth portion is the highest point of the second guide surface **431** in the axial direction L. Allowing the highest point of the first guide surface **331** in the axial direction L to come into contact with the highest point of the second guide surface **431** in the axial direction L enables the delivery tube holder **300** to be in the first position, that is, in the lowest position in the axial direction L. In addition, after the highest point of the first guide surface **331** is separated from the highest point of the second guide surface **431**, the radial flat portion **332-1** of the stop portion **332** of the first guide surface **331** stops the highest point of the second guide surface **431**, so as to temporarily stop the axial upward movement of the delivery tube holder **300** at the second position to enable the atomizer **1000** to perform pre-spraying before the formal spraying. In the embodiment described above, the second portion is the same as the fourth portion, both being the highest point of the second guide surface **431** in the axial direction L. In some other embodiments, the second portion and the fourth portion may be different portions. For example, the second portion is the highest point of the second guide surface **431** in the axial direction L, and the fourth portion is the second highest point of the second guide surface **431** in the axial direction L.
[0064] In some embodiments, as shown in FIGS. **4**D, **5**D and **6**D, the first guide surface **331** further includes a transition portion **334**. The transition portion **334** is located between the guide portion **333** and the stop portion **332** and is configured as a cambered surface. This facilitates the highest point of the second guide surface **431** to slide from the highest point of the first guide surface **331** to come into contact with the step surface of the first guide surface **331**.
[0065] In some embodiments, the width of the first guide surface **331** in the radial direction J of the delivery tube holder **300** is different from the width of the second guide surface **431** in the radial direction J. Specifically, the width of the first guide surface **331** in the radial direction J may be greater than or less than the width of the second guide surface **431** in the radial direction J. This can reduce the frictional force when the first guide surface **331** and the second guide surface **431** slide relatively, such that the user's operation is more labor-saving.
[0066] In some embodiments, as shown in FIGS. **7**A, **7**B, **8**A and **8**B, the first guide structure **330** on the inner wall of the delivery tube holder **300** may extend along an inner circumferential wall of the delivery tube holder **300**, and correspondingly, the second guide structure **430** on the inner wall of the mouthpiece **400** may extend around the central axis of the mouthpiece **400**.
[0067] In some embodiments, as shown in FIGS. **7**A, **7**B, **8**A and **8**B, two or more first guide structures **330** may be provided on the inner wall of the delivery tube holder **300**, and correspondingly, two or more second guide structures **430** may be provided on the inner wall of the mouthpiece **400**. The number of the first guide structures **330** may be the same as or different from the number of the second guide structures **430**. Thus, the atomizer **1000** can perform two or more stroke cycles of spraying when the delivery tube holder **300** rotates one circle, each stroke cycle of spraying including pre-spraying and formal drug-delivery spraying.
[0068] In some embodiments, the first guide structure **330** is integrally formed with the delivery tube holder **300**, and the second guide structure **430** is integrally formed with the mouthpiece **400**. This makes it easier to manufacture the first guide structure **330** and the second guide structure **430**.
[0069] In some embodiments, as shown in FIGS. **4**B, **5**B, **6**B and **10**, the atomizer **1000** may further include an actuating member **500**. The actuating member **500** can be actuated to drive the delivery tube holder **300** to pivot relative to the mouthpiece **400**, and the pivoted delivery tube holder **300** can be guided by the guide mechanism **440** to move in the axial direction L.
[0070] In some embodiments, as shown in FIG. **10**, the actuating member **500** may have a cylindrical structure, and may be sleeved on the delivery tube holder **300** so as to drive the delivery

tube holder **300** to rotate. Alternatively, the actuating member **500** may also be in other shapes, such as an oval shape. In addition, the actuating member **500** may be configured as the housing of the atomizer **1000**, or the actuating member **500** may be connected to a housing **100** of the atomizer **1000**, such that the delivery tube holder **300** can rotate or move in a spiral manner as the housing **100** rotates. The actuating member **500** may include an actuating member body **510**.

[0071] In some embodiments, as shown in FIG. **3**, the mouthpiece **400** may be sleeved on the nozzle assembly **800** and/or the delivery tube holder **300**, such that an atomized medicinal liquid is sprayed through one end of the mouthpiece **400**. The mouthpiece **400** may include a mouthpiece body **410**.

[0072] In some embodiments, as shown in FIGS. **4**B, **5**B, **6**B and **9**, the atomizer **1000** may further include a locking mechanism **530**. The locking mechanism **530** switches the actuating member **500** between a locked position, in which the mouthpiece **400** stops the actuating member **500** from pivoting, and an unlocked position, in which the actuating member **500** can pivot relative to the mouthpiece **400**.

[0073] In some embodiments, the locking mechanism **530** may include a first locking structure **420** arranged on the mouthpiece body **410** and a second locking structure **520** arranged on the actuating member body **510**. As shown in FIGS. **4**B, **5**B and **6**B, the first locking structure **420** may include a locking slider **421**, and correspondingly, the second locking structure **520** may include a slot **521** for accommodating the locking slider **421**. In this case, in the locked position, the locking slider **421** may be inserted into the slot **521** to stop the actuating member **500** from pivoting relative to the mouthpiece **400**, as shown in FIG. **5**B. In the implementation described above, the actuating member **500** can be effectively locked by means of a fixing member (i.e., the mouthpiece), which improves the locking reliability of the actuating member and thus avoids automatic trigger of the atomizer **1000** as much as possible. In some other embodiments, the first locking structure **420** may include a slot, and correspondingly, the second locking structure **520** may include a locking slider. Alternatively, the first locking structure **420** and the second locking structure **520** may also have other structures that can be selectively engaged and separated, such as a snap-fit structure, so as to lock or unlock the actuating member **500** relative to the mouthpiece **400**.

[0074] In some embodiments, as shown in FIGS. **4**B, **5**B and **6**B, the locking slider **421** can move relative to the mouthpiece body **410** in the axial direction L of the mouthpiece **400**, and the slot **521** extends in the axial direction L from an end surface of the actuating member body **510** facing the mouthpiece **400**. By means of providing the locking slider **421** that can move axially and the slot **521** that extends in the axial direction L as described above, the locking slider **421** can slide into the slot **521** by its own gravity when facing the slot **521**, so as to lock the actuating member **500**. Alternatively, the locking slider may also move in the radial direction J relative to the mouthpiece body **410**, and correspondingly, the slot extends from a circumferential surface of the actuating member body **510** in the radial direction J. The disclosure is not limited to the example described above.

[0075] In some embodiments, in the locked position, a portion of the locking slider **421** in the axial direction L is inserted into the slot **521** to stop the actuating member **500** from pivoting relative to the mouthpiece **400**; and in the unlocked position, the locking slider **421** abuts on the end surface of the actuating member body **510** facing the mouthpiece **400**. Specifically, in the unlocked position, the locking slider **421** on the mouthpiece body **410** abuts on the end surface of the actuating member body **510** facing the mouthpiece **400**. In this case, during the pivoting of the actuating member **500** relative to the mouthpiece **400**, the locking slider **421** may slide on the end surface of the actuating member body **510** facing the mouthpiece **400** until the locking slider **421** is about to directly face the slot **521** in the actuating member body **510**, as shown in FIG. **4**B. Then, the actuating member **500** continues to be pivoted to allow the locking slider **421** to directly face the slot **521**, and at this time, a portion of the locking slider **421** slides into the slot **521** of the actuating member body **510**, such that the mouthpiece **400** and the actuating member **500** stop each other, as shown in FIG. **5**B.

[0076] In some embodiments, as shown in FIG. **9**, the locking slider **421** may have a T-shaped cross-section. The disclosure is not limited thereto. In addition, a hole **423** may be provided in the locking slider **421** for accommodating an elastic member **422** (described in detail below).

[0077] In some embodiments, as shown in FIG. **10**, a side wall of the slot **521** may be provided with an inclined surface that is inclined from the side wall toward the end surface of the actuating member body **510** facing the mouthpiece **400**. This makes it easy for the locking slider **421** to slide into the slot **521** when facing the slot **521**, and also makes it easy for the locking slider **421** to return to and abut on the end surface of the actuating member body **510** facing the mouthpiece **400** after the locking slider **421** returns to the mouthpiece body **410**.

[0078] In some embodiments, as shown in FIGS. **3**, **8**A and **8**B, the mouthpiece body **410** may include a recess **411** for accommodating the locking slider **421**, and the first locking structure **420** further includes the elastic member **422**. One end of the elastic member **422** is fixed to an inner wall of the recess **411**, and the other end of the elastic member **422** is fixed to the locking slider **421**. The locking slider **421** may slide in the recess **411** of the mouthpiece body **410**. When the actuating member **500** is in the locked position, a portion of the locking slider **421** may remain in the recess **411** of the mouthpiece body **410**, and the other portion of the locking slider is inserted into the slot **521** of the actuating member body **510**, such that the mouthpiece **400** and the actuating member **500** stop each other. In addition, providing the elastic member may further promote the movement of the locking slider **421**. In some examples, the elastic member may be a spring, etc. Alternatively, the first locking structure **420** may include no elastic member. In this case, when the locking slider **421** faces the slot **521**, the locking slider **421** may slide into the slot **521** only by its own gravity. In some other embodiments, the first locking structure **420** may further include other drive devices, such as an electric motor, to control the axial movement of the locking slider **421**.

[0079] In some embodiments, when the actuating member **500** is in the unlocked position, the elastic member **422** is in a compressed state; and when the actuating member **500** is switched from the unlocked position to the locked position, the elastic member **422** returns to enable the locking slider **421** to enter the slot **521**. The elastic force of the elastic member **422** may promote the locking slider **421** to be inserted into the slot **521** quickly and accurately, thereby improving the locking reliability of the actuating member **500**.

[0080] In some embodiments, as shown in FIGS. **5**B, **7**A and **7**B, a protrusion **310** is provided on the circumferential surface of the delivery tube holder **300**. The protrusion **310** extends into the slot **521**. In the locked position, the locking slider **421** abuts on the protrusion **310**, such that when the delivery tube holder **300** moves toward the mouthpiece, the protrusion **310** pushes the locking slider **421** to return to the mouthpiece body **410**. Specifically, as shown in FIG. **5**B, in the locked position, the locking slider **421** is inserted into the slot **521** of the actuating member body **510** and abuts on the protrusion **310** of the delivery tube holder **300**, so as to lock the actuating member **500**. Then, for example, a switch button **900** can be pressed to enable the delivery tube holder **300** to continue to rotate, such that the delivery tube holder **300** can move toward the mouthpiece **400** under the action of the bottom spring **600** and drive, via the protrusion **310**, the locking slider **421** to return into the mouthpiece body, as shown in FIG. **6**B. At this time, the actuating member **500** returns to the unlocked position, the actuating member **500** continues to rotate to enable the locking slider **421** to abut on the end surface of the actuating member body **510** facing the mouthpiece again, and thus the atomizer **1000** returns to the initial state and is ready for the next spraying operation (dosing cycle).

[0081] In some embodiments, as shown in FIG. **4**B, in the unlocked position, the protrusion **310** comes into contact with the slot **521**, such that the actuating member **500** can push the delivery tube holder **300** to pivot via the protrusion **310**. According to the implementation described above, the protrusion **310** on the delivery tube holder **300** not only enables the locking slider **421** to return to the mouthpiece, but also serves as a transmission member to transmit the rotational motion of the actuating member **500** to the delivery tube holder **300**, thereby simplifying the structure of the atomizer **1000** to make the atomizer **1000** more compact and reduce the cost. In some other embodiments, the actuating member may drive the delivery tube holder **300** to rotate by means of a transmission member having other structures, for example, a protrusion provided on the actuating member body **510**.

[0082] In some embodiments, when the delivery tube holder **300** is in the first position and the third position, the actuating member **500** is in the unlocked position, as shown in FIGS. **4**A, **4**B, **6**A and **6**B; and when the delivery tube holder **300** is in the second position, the actuating member **500** is in the locked position, as shown in FIGS. **5**A and **5**B. This can prevent the atomizer **1000** from being automatically triggered, after entering the cleaned state (i.e., the pre-triggering state), due to the user continuing to rotate the actuating member **500**. Specifically, when the delivery tube holder **300** is in the first position, the locking slider **421** abuts against the end surface of the actuating member **500** facing the mouthpiece **400**, such that the actuating member **400** is in the unlocked position, as shown in FIGS. **4**A and **4**B. When the delivery tube holder **300** is in the second position, a portion of the locking slider **421** is inserted into the slot **521** and abuts on the protrusion **310**, such that the actuating member **500** is in the locked position, as shown in FIGS. **5**A and **5**B. When the delivery tube holder **300** is in the third position, the locking slider is pushed by the protrusion **310** to return to the mouthpiece **400**, such that the actuating member **500** is in the unlocked position, as shown in FIGS. **6**A and **6**B.

[0083] In some embodiments, as shown in FIGS. **5**C, **6**C and **11**, the atomizer **1000** may further include a switch button **900**. In the locked position, the switch button **900** can be pressed to drive the delivery tube holder **300** to pivot to separate the first guide surface **331** from the second guide surface **431**, so as to enable the delivery tube holder **300** to move from the second position to the third position under the action of the bottom spring **600**, such that the atomizer **1000** enters the triggered state from the pre-triggering state (i.e., the cleaned state).

[0084] In some embodiments, as shown in FIGS. **4**C, **5**C, **6**C and **11**, a first inclined surface **911** is provided at an end portion **910** of the switch button **900** facing the delivery tube holder **300**, the first inclined surface **911** is configured to be inclined from the end **910** away from the delivery tube holder **300**, an opening **320** is provided in a circumferential wall of the delivery tube holder **300**, and the opening **320** is configured to accommodate the end portion **910**. Thus, when the switch button **900** is pressed, the first inclined surface **911** on the switch button **900** can push a side wall on the opening **320** of the delivery tube holder **300** so as to push the delivery tube holder **300** to pivot.

[0085] In some embodiments, as shown in FIGS. **5**C, **6**C, **7**B and **11**, a third inclined surface **912** opposite to the first inclined surface **911** is further provided at the end portion of the switch button **900**. The third inclined surface **912** is configured to be inclined from the end portion **910** away from the delivery tube holder **300**, which can reduce resistance to make it easy for a second inclined surface **323** (described in detail below) of the delivery tube holder **300** to come into contact with the end portion **910** of the switch button **900**.

[0086] In some embodiments, as shown in FIG. **4**C, in the unlocked position, the end portion **910** faces the circumferential wall of the delivery tube holder **300**. This can ensure that the switch button **900** cannot be pressed when the atomizer **1000** does not enter the pre-triggering state, thereby ensuring normal drug delivery of the atomizer **1000**.

[0087] In some embodiments, as shown in FIG. **5**C, in the locked position, the end portion **910** faces a first side wall **321** of the opening **320**, and when the switch button **900** is pressed, the first inclined surface **911** can come into contact with the first side wall **321** of the opening **320** to push the delivery tube holder **300** to pivot. With the arrangement in which the end portion **910** of the switch button **900** faces the first side wall **321** of the opening **320** of the delivery tube holder **300** when in the locked position, when the switch button **900** is pressed, the end portion **910** of the switch button **900** can enter the opening **320** to enable the first inclined surface **911** to come into contact with the first side wall **321** of the opening **320**. In this way, the switch button **900** is used to drive the delivery tube holder **300** to rotate to enable the atomizer **1000** to enter the triggered state from the pre-triggering state.

[0088] In some embodiments, as shown in FIGS. **4**C, **5**C, **6**C and **7**A, the second inclined surface **323** inclined from a second side wall **322** toward the switch button **900** is provided on the second side wall **322** of the opening **320** facing the first side wall **321**; and as shown in FIG. **6**C, when the atomizer **1000** is in the triggered state, the actuating member **500** continues to be pivoted to cause the second inclined surface **323** to come into contact with the end portion **910**, so as to enable the switch button **900** to move away from the delivery tube holder **300**. Specifically, after the switch button **900** is pressed to enable the atomizer **1000** to enter the triggered state, the actuating member **500** returns to the unlocked position, and at this time, the actuating member **500** may be rotated to drive the delivery tube holder **300** to rotate, so as to cause the second inclined surface **323** of the opening **320** of the delivery tube holder **300** to come into contact with the end portion **910** of the switch button **900**. In this way, the rotation of the delivery tube holder **300** can push the switch button **900** to return to the initial state to be ready for the next spraying operation.

[0089] To this end, the actuating member **500** can drive the delivery tube holder **300** to rotate in the unlocked position to switch the atomizer **1000** from the initial state to the pre-cleaning state and from the pre-cleaning state to the cleaned state and the pre-triggering state, and the switch button **900** can be pressed in the locked position to push the delivery tube holder **300** to continue to rotate to enable the atomizer **1000** to enter the triggered state from the pre-triggering state. After the atomizer **1000** enters the triggered state, the actuating member **500** also returns to the unlocked position. At this time, the actuating member **500** can be rotated to drive the delivery tube holder **300** to rotate, so as to enable the atomizer **1000** to return to the initial state.

[0090] In some embodiments, the atomizer **1000** may further include a bottom spring **600** that is arranged at the bottom of the delivery tube holder **300** and configured to apply a thrust to the delivery tube holder **300** to drive the delivery tube holder **300** to move toward the mouthpiece **400**. Specifically, when the first guide surface **331** is separated from the second guide surface **431**, the delivery tube holder **300** can move from the second position to the third position under the action of the thrust of the bottom spring **600**. In this way, the delivery tube holder **300** moves from the first position to the second position and the third position under the combined action of the guide mechanism **440** and the bottom spring **600**, so as to enable the atomizer **1000** to perform the pre-spraying and the formal drug-delivery spraying.

[0091] In some embodiments, the atomizer **1000** may further include other members, such as a housing **100**, a container **700** for holding the liquid to be atomized, a spraying assembly **800** for atomizing the liquid and spraying the atomized liquid, and a dust cover **200**.

[0092] According to some embodiments of the disclosure described above, the atomizer **1000** may be, but does not necessarily, operated as follows.

[0093] When the atomizer **1000** is in the initial state, the first guide structure **330** on the inner wall of the delivery tube holder **300** is separated from the second guide structure **430** on the inner wall of the mouthpiece **400**, the locking slider **421** on the mouthpiece **400** abuts on the end surface of the actuating member body **510** facing the mouthpiece **400**, and the end portion **910** of the switch button **900** faces the circumferential wall of the delivery tube holder **300**.

[0094] From the initial state, the user holds the housing **100** and rotates the housing **100** by an angle, and the actuating member **500** drives the delivery tube holder **300** to rotate along with the housing **100**, such that the first guide structure **330** on the delivery tube holder **300** comes into contact with the second guide structure **430** on the mouthpiece **400**. At this time, the delivery tube holder **300** may move downward in a spiral manner under the action of the first guide structure **330** and the second guide structure **430**. As the actuating member **500** rotates, the slot **521** in the actuating member **500** approaches the locking slider **421** on the mouthpiece **400**, as shown in FIG. **4**B, and the end portion **910** of the switch button **900** also approaches the first side wall **321** of the opening **320** of the delivery tube holder **300**, as shown in FIG. **4**C. The actuating member **500** continues to rotate until the highest point of the first guide structure **330** comes into contact with the highest point of the second guide structure **430**, and at this time, the delivery tube holder **300** is in the first position, that is, the atomizer **1000** enters the pre-cleaning state, as shown in FIG. **4**A.

[0095] When the atomizer **1000** is in the pre-cleaning state, the bottom spring **600** of the delivery tube holder **300** is compressed to the maximum extent, as shown in FIG. **4**A. At this time, the actuating member **500** continues to rotate, the highest point of the first guide structure **330** is separated from the highest point of the second guide structure **430**, the delivery tube holder **300** moves toward the mouthpiece **400** in the axial direction L under the action of the upward thrust of the bottom spring **600** until the step surface (specifically, the radial flat portion) of the first guide structure **330** comes into contact with the highest point of the second guide structure **430**, and at this time, the delivery tube holder **300** is in the second position, that is, the atomizer **1000** enters the cleaned state, as shown in FIG. **5**A. The process described above is a pre-spraying process of the atomizer.

[0096] When the atomizer **1000** is in the cleaned state (also referred to as a pre-triggering state), a portion of the locking slider **421** on the mouthpiece **400** is inserted into the slot **521** of the actuating member body **510** and abuts on the protrusion **310** on the circumferential wall of the delivery tube holder **300**, such that the actuating member **500** is in the locked position, as shown in FIG. **5**B. In addition, the end portion **910** of the switch button **900** faces the first side wall **321** of the delivery tube holder **300**, as shown in FIG. **5**C. In the locked position, if the user does not apply an external force by means of other components, the atomizer **1000** does not perform an atomization or spraying operation, even if the atomizer **1000** is subjected to an external force, such as impact or shaking, or even accidents such as slipping from a height. Thus, the atomizer **1000** is effectively locked in this state.

[0097] When the user wants to use the atomizer **1000** to perform formal drug-delivery spraying, the user may apply a radial inward thrust to the switch button **900**, that is, press the switch button **900**. The switch button **900** is pressed such that the first inclined surface **911** on the switch button **900** comes into contact with the first side wall **321** of the opening **320** of the delivery tube holder **300** and pushes the first side wall **321** to move, so as to drive the delivery tube holder **300** to further rotate. At this time, the protrusion **310** on the delivery tube holder **300** also moves a distance in the slot **521** along with the delivery tube holder **300**. When the second guide surface **431** of the second guide structure **430** of the mouthpiece **400** is separated from the first guide surface **331** of the first guide structure **330** of the delivery tube holder **300**, the delivery tube holder **300** moves toward the mouthpiece **400** in the axial direction L under the action of the upward thrust of the bottom spring **600** to enable the atomizer **1000** to enter the triggered state, as shown in FIG. **6**A. In addition, the protrusion **310** on the delivery tube holder **300** pushes the locking slider **421** to return into the mouthpiece body, so as to enable the actuating member **500** to return to the unlocked position, as shown in FIG. **6**B.

[0098] Then, the user continues to rotate the housing **100** by an angle to enable the actuating member **500** to continue to drive the delivery tube holder **300** to rotate along with the housing **100**. The locking slider **421** returns to and abuts on the end surface of the actuating member body **510** facing the mouthpiece **400**. In addition, as shown in FIG. **6**C, the second inclined surface **323** of the second side wall **322** of the delivery tube holder **300** comes into contact with the end portion **910** of the switch button **900** and pushes the switch button **900** to return to the original position, such that the atomizer **1000** returns to the initial position. At this point, the atomizer **1000** completes one stroke cycle of spraying.

[0099] An atomizer **1000′** according to some specific embodiments of the disclosure in which no pre-spraying is performed will be described in detail below with reference to the embodiments shown in FIGS. **12**A to **12**C. It should be noted herein that, except the feature that no limiting step for pre-spraying is provided on a first guide surface of a first guide structure **330′** in the

embodiments in which no pre-spraying is performed, other features of the delivery tube holder of the atomizer **1000**′ in the embodiments in which no pre-spraying is performed and the features of other components such as a nozzle, an actuating member, a locking mechanism and a switch button are the same as the features of the delivery tube holder of the atomizer **1000** in the embodiments in which pre-spraying is performed and the features of the components such as the nozzle, the actuating member, the locking mechanism and the switch button, and will not be described in detail herein for the sake of brevity. Only the differences between the atomizer **1000**′ in the embodiments in which no pre-spraying is performed and the atomizer **1000** in the embodiments in which pre-spraying is performed will be described in detail below.
[0100] In some embodiments, when the delivery tube holder **300** is in the first position, the atomizer **1000**′ is in a pre-triggering state, as shown in FIG. **12**A. When the delivery tube holder **300** is in the third position, the atomizer **1000**′ is in a triggered state or an initial state, as shown in FIG. **13**A. Thus, the atomizer **1000**′ is guided by the guide mechanism **440**′ to perform formal drug-delivery spraying and to be ready for the next drug-delivery spraying within one stroke cycle. In some embodiments, the guide mechanism **440**′ may include: a first guide structure **330**′ arranged on an inner wall of the delivery tube holder **300**, with a surface of the first guide structure **330**′ facing the mouthpiece **400** forming the first guide surface; and a second guide structure **430** arranged on an inner wall of the mouthpiece **400**, with a surface of the second guide structure **430** facing the delivery tube holder **300** forming a second guide surface. Thus, during the pivoting of the delivery tube holder **300**, the delivery tube holder **300** can move in the axial direction under the interaction of the first guide surface and the second guide surface that are in contact with each other, that is, can perform a composite motion of rotation and axial translation, so as to change the position of the delivery tube holder **300** in the axial direction L.
[0101] In some embodiments, during the pivoting of the delivery tube holder **300**, the delivery tube holder **300** may move downward while rotating under the action of the first guide surface and the second guide surface, and during the upward movement of the delivery tube holder **300**, the delivery tube holder **300** may perform a simple upward translation without any rotation. The disclosure is not limited thereto. When the delivery tube holder is in the first position, the highest point of the first guide surface in the axial direction comes into contact with the highest point of the second guide surface in the axial direction, as shown in FIG. **12**A, and at this time, the delivery tube holder is farthest from the mouthpiece. Then, after the first guide surface is separated from the second guide surface, the delivery tube holder moves to the third position.
[0102] In some embodiments, the first guide surface may include an inclined portion and a flat portion. The inclined portion may be a spiral surface, for example, a spiral surface that spirally extends upward from an inner bottom wall of the delivery tube holder **300** around a central axis of the delivery tube holder **300**. Alternatively, the inclined portion may also be an inclined surface or a curved surface that extends in any direction and has a height gradually increasing from the inner bottom wall of the delivery tube holder **300**. The disclosure is not limited to the example described above. The flat portion may be in the highest position of the first guide surface in the axial direction L.
[0103] In some embodiments, the second guide surface **431** may also include an inclined portion and a flat portion. The inclined portion may be a spiral surface, for example, a spiral surface extending upward from an inner bottom wall of the mouthpiece **400** around a central axis of the mouthpiece **400**. Alternatively, the inclined portion may also be an inclined surface or a curved surface that extends in any direction and has a height gradually increasing from the inner bottom wall of the mouthpiece **400**. The disclosure is not limited to the example described above. The flat portion may be in the highest position of the second guide surface in the axial direction L.
[0104] In some embodiments, when the delivery tube holder **300** is in the first position, the actuating member **500** is in an unlocked position (as shown in FIGS. **12**A and **12**B); and when the delivery tube holder **300** is in the third position, the actuating member **500** is in a locked position,

as shown in FIGS. **13**A and **13**B. When the delivery tube holder **300** is in the first position, a locking slider **421** may be inserted into a slot **521** to stop the actuating member **500** from pivoting relative to the mouthpiece **400**, so as to enable the actuating member to be in the locked position, as shown in FIG. **12**B. In the implementation described above, the actuating member **500** can be effectively locked by means of a fixing member (i.e., the mouthpiece), which improves the locking reliability of the actuating member and thus avoids automatic trigger of the atomizer **1000′** as much as possible. The locking slider **421** can move relative to the mouthpiece body in the axial direction L of the mouthpiece **400**, and the slot **521** extends in the axial direction L from an end surface of the actuating member body **510** facing the mouthpiece **400**. By means of providing the locking slider **421** that can move axially and the slot **521** that extends in the axial direction L as described above, the locking slider **421** can slide into the slot **521** by its own gravity when facing the slot **521**, so as to lock the actuating member **500**.

[0105] In addition, when the delivery tube holder **300** is in the first position, a portion of the locking slider **421** is inserted into the slot **521** and abuts on a protrusion **310**, such that the actuating member **500** is in the locked position, as shown in FIGS. **12**A and **12**B. When the delivery tube holder **300** is in the third position, the locking slider is pushed by the protrusion **310** to return to the mouthpiece **400**, such that the actuating member **500** is in the unlocked position, as shown in FIGS. **13**A and **13**B.

[0106] In some embodiments, as shown in FIG. **12**C, in the locked position, the switch button **900** can be pressed to drive the delivery tube holder **300** to pivot to separate the first guide surface from the second guide surface, so as to enable the delivery tube holder **300** to move from the first position to the third position under the action of a bottom spring **600**, such that the atomizer **1000′** enters the triggered state from the pre-triggering state. As shown in FIG. **13**C, when the atomizer **1000′** is in the triggered state, the actuating member **500** continues to be pivoted to cause a second inclined surface **323** to come into contact with an end portion **910**, so as to enable the switch button **900** to move away from the delivery tube holder **300**. Specifically, after the switch button **900** is pressed to enable the atomizer **1000′** to enter the triggered state, the actuating member **500** returns to the unlocked position, and at this time, the actuating member **500** may be rotated to drive the delivery tube holder **300** to rotate, so as to cause the second inclined surface **323** of the opening **320** of the delivery tube holder **300** to come into contact with the end portion **910** of the switch button **900**. In this way, the rotation of the delivery tube holder **300** can push the switch button **900** to return to the initial state to be ready for the next spraying operation.

[0107] According to some embodiments of the disclosure described above, the atomizer **1000′** may be, but does not necessarily, operated as follows.

[0108] When the atomizer **1000′** is in the initial state, the first guide structure **330′** on the inner wall of the delivery tube holder **300** is separated from the second guide structure **430** on the inner wall of the mouthpiece **400**, the locking slider **421** on the mouthpiece **400** abuts on the end surface of the actuating member body **510** facing the mouthpiece **400**, and the end portion **910** of the switch button **900** faces the circumferential wall of the delivery tube holder **300**.

[0109] From the initial state, the user holds the housing and rotates the housing by an angle, and the actuating member **500** drives the delivery tube holder **300** to rotate along with the housing, such that the first guide structure **330′** on the delivery tube holder **300** comes into contact with the second guide structure **430** on the mouthpiece **400**. At this time, the delivery tube holder **300** may move downward in a spiral manner under the action of the first guide structure **330′** and the second guide structure **430**. As the actuating member **500** rotates, the slot **521** in the actuating member **500** approaches the locking slider **421** on the mouthpiece **400**, and the end portion **910** of the switch button **900** also approaches the first side wall **321** of the opening **320** of the delivery tube holder **300**, as shown in FIG. **12**C. The actuating member **500** continues to rotate until the highest point of the first guide structure **330′** comes into contact with the highest point of the second guide structure **430**, and at this time, the delivery tube holder **300** is in the first position, that is, the atomizer **1000** enters the pre-triggering state, as shown in FIG. **12**A, and the locking slider **421** slides into the slot **521** to lock the actuating member **500**, as shown in FIG. **12**B.

[0110] When the atomizer **1000**′ is in the pre-triggering state, the bottom spring **600** of the delivery tube holder **300** is compressed to the maximum extent, as shown in FIG. **12**A. At this time, when the user wants to use the atomizer **1000**′ to perform formal drug-delivery spraying, the user may apply a radial inward thrust to the switch button **900**, that is, press the switch button **900**. The switch button **900** is pressed such that the first inclined surface **911** on the switch button **900** comes into contact with the first side wall **321** of the opening **320** of the delivery tube holder **300** and pushes the first side wall **321** to move, so as to drive the delivery tube holder **300** to further rotate. At this time, the protrusion **310** on the delivery tube holder **300** also moves a distance in the slot **521** along with the delivery tube holder **300**. When the second guide surface of the second guide structure **430** of the mouthpiece **400** is separated from the first guide surface of the first guide structure **330** of the delivery tube holder **300**, the delivery tube holder **300** moves toward the mouthpiece **400** in the axial direction L under the action of the upward thrust of the bottom spring **600** to enable the atomizer **1000**′ to enter the triggered state, as shown in FIG. **13**A. In addition, the protrusion **310** on the delivery tube holder **300** pushes the locking slider **421** to return into the mouthpiece body, so as to enable the actuating member **500** to return to the unlocked position, as shown in FIG. **13**B.

[0111] Then, the user continues to rotate the housing by an angle to enable the actuating member **500** to continue to drive the delivery tube holder **300** to rotate along with the housing. The locking slider **421** returns to and abuts on the end surface of the actuating member body **510** facing the mouthpiece **400**. In addition, as shown in FIG. **13**C, the second inclined surface **323** of the second side wall **322** of the delivery tube holder **300** comes into contact with the end portion **910** of the switch button **900** and pushes the switch button **900** to return to the original position, such that the atomizer **1000**′ returns to the initial position. At this point, the atomizer **1000**′ completes one stroke cycle of spraying.

Claims

1. An atomizer, comprising: a mouthpiece, a first end of the mouthpiece being configured to be placed at a mouth of a user; a delivery tube holder located at a second end of the mouthpiece opposite to the first end; and a guide mechanism configured to guide the delivery tube holder to move between a first position and a third position in an axial direction of the mouthpiece, wherein the delivery tube holder is closest to the mouthpiece in the axial direction when the delivery tube holder is in the third position, and the delivery tube holder is farthest from the mouthpiece in the axial direction when the delivery tube holder is in the first position; and wherein the atomizer performs spraying during the movement of the delivery tube holder from the first position to the third position, and the atomizer performs liquid metering during the movement of the delivery tube holder from the third position to the first position.

2. The atomizer according to claim 1, wherein there is a second position between the first position and the third position, and the guide mechanism is configured to guide the delivery tube holder to move between the first position, the second position and the third position in the axial direction of the mouthpiece; and wherein the atomizer performs liquid metering during the movement of the delivery tube holder from the third position to the first position, the atomizer performs a first spraying operation during the movement of the delivery tube holder from the first position to the second position, and the atomizer performs a second spraying operation during the movement of the delivery tube holder from the second position to the third position, so as to complete one dosing cycle.

3. The atomizer according to claim 2, wherein the guide mechanism comprises: a first guide structure arranged on an inner wall of the delivery tube holder, with a surface of the first guide

structure facing the mouthpiece forming a first guide surface; and a second guide structure arranged on an inner wall of the mouthpiece, with a surface of the second guide structure facing the delivery tube holder forming a second guide surface.
4. The atomizer according to claim 3, wherein when a first portion of the first guide surface abuts on a second portion of the second guide surface, the delivery tube holder is in the first position; when a third portion of the first guide surface abuts on a fourth portion of the second guide surface, the delivery tube holder is in the second position; and when the first guide surface is separated from the second guide surface, the delivery tube holder is in the third position.
5. The atomizer according to claim 4, wherein the first guide surface comprises: a guide portion configured to guide the delivery tube holder to move from the third position to the first position; and a stop portion configured to guide the delivery tube holder to move from the first position to the second position and stop the delivery tube holder from moving from the second position to the first position.
6. The atomizer according to claim 5, wherein the guide portion is an inclined surface, and the stop portion is a step surface and comprises an axial flat portion parallel to the axial direction and a radial flat portion parallel to a radial direction of the delivery tube holder.
7. The atomizer according to claim 6, wherein the first portion is the highest point of the guide portion in the axial direction, and the second portion is the highest point of the second guide surface in the axial direction; and wherein the third portion is the radial flat portion of the stop portion, and the fourth portion is the same portion as the second portion.
8. The atomizer according to claim 5, wherein the first guide surface further comprises a transition portion located between the guide portion and the stop portion and configured as a cambered surface.
9. The atomizer according to claim 3, wherein a width of the first guide surface in a radial direction of the delivery tube holder is different from a width of the second guide surface in the radial direction.
10. The atomizer according to claim 3, wherein the first guide structure is integrally formed with the delivery tube holder, and the second guide structure is integrally formed with the mouthpiece.
11. The atomizer according to claim 4, further comprising an actuating member, wherein the actuating member is capable of being actuated to drive the delivery tube holder to pivot relative to the mouthpiece, and the pivoted delivery tube holder is capable of being guided by the guide mechanism to move in the axial direction.
12. The atomizer according to claim 11, further comprising a locking mechanism, wherein the locking mechanism switches the actuating member between a locked position, in which the mouthpiece stops the actuating member from pivoting, and an unlocked position, in which the actuating member is capable of pivoting relative to the mouthpiece.
13. The atomizer according to claim 12, wherein when the delivery tube holder is in the first position and the third position, the actuating member is in the unlocked position; and when the delivery tube holder is in the second position, the actuating member is in the locked position.
14. The atomizer according to claim 12, further comprising a switch button, wherein in the locked position, the switch bottom is capable of being pressed to drive the delivery tube holder to pivot to separate the first guide surface from the second guide surface.
15. The atomizer according to claim 12, further comprising a bottom spring arranged at the bottom of the delivery tube holder, wherein the bottom spring is configured to apply a thrust to the delivery tube holder to drive the delivery tube holder to move toward the mouthpiece, and when the first guide surface is separated from the second guide surface, the delivery tube holder is capable of moving from the second position to the third position under the action of the thrust of the bottom spring.
16. The atomizer according to claim 12, wherein the locking mechanism comprises: a locking slider provided on the mouthpiece; a slot provided in a circumferential wall of the actuating

member; and a protrusion provided on a circumferential surface of the delivery tube holder and extending into the slot, wherein when the delivery tube holder is in the first position, the locking slider abuts against an end surface of the actuating member facing the mouthpiece, such that the actuating member is in the unlocked position; wherein when the delivery tube holder is in the second position, a portion of the locking slider is inserted into the slot and abuts on the protrusion, such that the actuating member is in the locked position; and wherein when the delivery tube holder is in the third position, the locking slider is pushed by the protrusion to return to the mouthpiece, such that the actuating member is in the unlocked position.

17. The atomizer according to claim 2, wherein when the delivery tube holder is in the first position, the atomizer is in a pre-cleaning state; when the delivery tube holder is in the second position, the atomizer is in a cleaned state and a pre-triggering state; and when the delivery tube holder is in the third position, the atomizer is in a triggered state or an initial state.

18. The atomizer according to claim 12, wherein when the delivery tube holder is in the first position, the actuating member is in the locked position; and when the delivery tube holder is in the third position, the actuating member is in the unlocked position.

19. The atomizer according to claim 12, wherein the locking mechanism comprises: a locking slider provided on the mouthpiece; a slot provided in a circumferential wall of the actuating member; and a protrusion provided on a circumferential surface of the delivery tube holder and extending into the slot, wherein when the delivery tube holder is in the first position, a portion of the locking slider is inserted into the slot and abuts on the protrusion, such that the actuating member is in the locked position; and wherein when the delivery tube holder is in the third position, the locking slider is pushed by the protrusion to return to the mouthpiece, such that the actuating member is in the unlocked position.